

Networks

Flows

Michel Bierlaire

Optimization: principles and algorithms



Flows

Physical networks are often used to transport objects or information: water, electricity, cars, internet packets, etc.

We provide a generic representation of flows of these objects, and associate them with the graph.

In our mathematical formalism, a directed graph is called a network when its nodes and arcs are associated with quantities.

The first of the quantities is the flow.

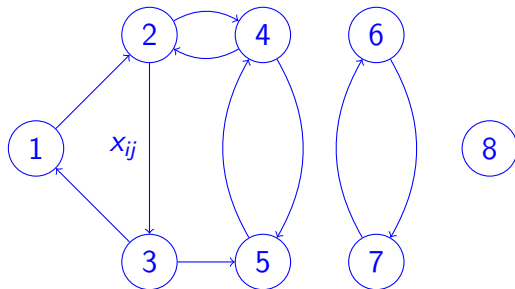
Flows



Definition

$$x_{ij} \in \mathbb{R}$$

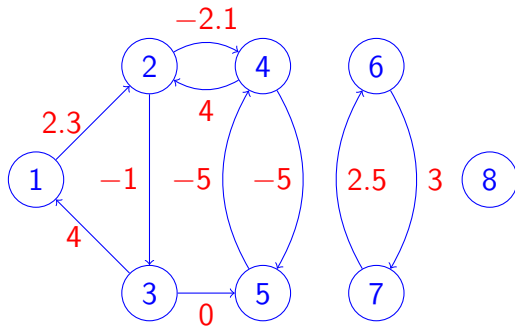
- ▶ Amount of “things” traversing the arc during a given time period.
- ▶ Associated with each arc (i, j) .
- ▶ Units are arbitrary and context dependent.
- ▶ Time period is irrelevant and long enough.



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Flow through a cut

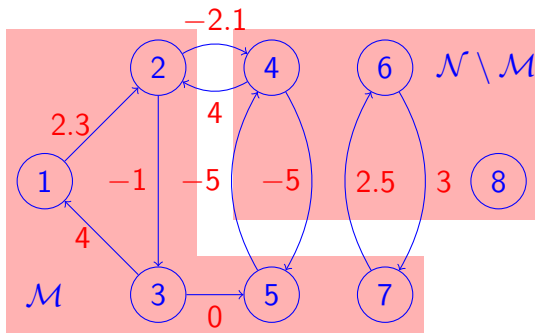
$$X(\Gamma) = \sum_{(i,j) \in \Gamma \rightarrow} x_{ij} - \sum_{(i,j) \in \Gamma \leftarrow} x_{ij},$$

$$\begin{aligned} \sum_{(i,j) \in \Gamma \rightarrow} x_{ij} = \\ x_{24} + x_{54} + x_{76} = -2.1 - 5 + 2.5 = -4.6 \end{aligned}$$

$$\begin{aligned} \sum_{(i,j) \in \Gamma \leftarrow} x_{ij} = \\ x_{42} + x_{45} + x_{67} = 4 - 5 + 3 = 2. \end{aligned}$$

Total flow through the cut:

$$-4.6 - 2 = -6.6.$$



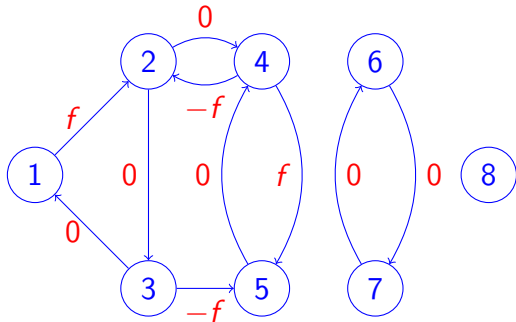
Simple path flow

$x \in \mathbb{R}^n$ such that

$$x_{ij} = \begin{cases} f & \text{if } (i,j) \in P^{\rightarrow} \\ -f & \text{if } (i,j) \in P^{\leftarrow} \\ 0 & \text{otherwise.} \end{cases}$$

Example

$$1 \rightarrow 2 \leftarrow 4 \rightarrow 5 \leftarrow 3$$



Summary

The concept of flow allows to model the transportation of things through the network

It is defined on each arc, but can be also interpreted with cuts and paths.

Negative flows represent backward movements.